Shiro Ikeda

Curriculum Vitae

10–3 Midoricho
Tachikawa, Tokyo, 190-8562
Japan
+81(0)50 5533 8426
ikeda46@gmail.com
ikeda46.github.io
0000-0002-2462-1448
Shiro-Ikeda
8cqUEh4AAAAJ

Personal Details

Gender:	Male
Date of birth:	21/December/1968
Place of birth:	Tokyo, Japan
Language:	Japanese and English
Citizenship:	Japan

Current Position

2016.04 – **Professor**, *The Institute of Statistical Mathematics*, Tokyo, Japan

2016.04 – **Professor**, *Statistical Science Program, The Graduate University of Advanced Studies (SOKENDAI)*, Tokyo, Japan

2017.04 – **Visiting Professor**, *National Astronomical Observatory of Japan*, Tokyo, Japan

2016.05 – **Visiting Senior Scientist**, *Kavli IPMU, The University of Tokyo*, Chiba, Japan

Past Academic Positions

2003.02 – 2016.03 **Associate Professor**, *The Institute of Statistical Mathematics*, Tokyo, Japan

2001.04 – 2003.01 **Associate Professor**, *Dept. of Brain Science and Eng., Kyushu Institute of Technology*, Fukuoka, Japan

1998.10 – 2001.03 **Researcher**, *"Information and Human Activity," PRESTO, JST*, Saitama, Japan

1996.04 – 1998.09 **Special Postdoctoral Researcher**, *RIKEN*, Saitama, Japan

1996.01 – 1996.03 **Research Fellow**, *JSPS (PD)*, Tokyo, Japan

Visiting Positions

2024.06 **Visiting Academic**, *GIPSA-Lab*, Grenoble, France

2022.05 – 2022.07 **Theoretical Science Visiting Program**, *OIST*, Okinawa, Japan

2010.04 – 2016.03 **Adjunct Associate Professor**, *Tokyo Institute of Technology*, Kanagawa, Japan

2008.04 – 2008.10 **Visiting Academic**, *The University of Melbourne*, Melbourne, Victoria, Australia

2007.10 – 2008.04 **Visiting Academic**, *Australian National University*, Canberra, ACT, Australia

2004.04 – 2006.03 **Visiting Researcher**, *RIKEN*, Saitama, Japan

2003.04 – 2004.03 **Visiting Academic**, *Gatsby Computational Neuroscience Unit, UCL*, London, UK, under the fellowship between JSPS and the Royal Society

Education

1993.04 – 1996.03 **Doctor of Engineering**, *The University of Tokyo*, Tokyo, Japan

Major Mathematical Engineering and Information Physics

Thesis Title	Estimating the structure of the sources
Supervisor	Kaoru Nakano
1991.04 – 1993.03	**Master of Engineering**, *The University of Tokyo*, Tokyo, Japan
Major	Mathematical Engineering and Information Physics
Thesis Title	Construct the structure of Hidden Markov Models
Supervisor	Kaoru Nakano
1987.04 – 1991.03	**Bachelor of Engineering**, *The University of Tokyo*, Tokyo, Japan
Major	Mathematical Engineering and Information Physics

Academic Honors and Awards

2025.07 **The International Congress for Basic Science 2025 Frontiers of Science Award in Theoretical Computer & Information Sciences**,
Recipients: The Event Horizon Telescope (EHT) Collaboration

2021.01 **Royal Astronomical Society 2021 Group Achievement Award**,
Recipients: The EHT Collaboration

2020.05 **2020 Einstein Medal**,
Recipients: The EHT Collaboration

2019.09 **2020 Breakthrough Prize in Fundamental Physics**,
Recipients: The EHT Collaboration

2019.05 **National Science Foundation Diamond Achievement Award**,
Recipients: The EHT Collaboration

2001.09 **Japan Neural Network Society Best Paper Award**,
S. Ikeda and K. Toyama "Independent component analysis for noisy data–MEG data analysis," Neural Networks, 13(10), 2000

Invited Talks (talks in English)

2025.06.06 **The imaging of the supermassive black hole shadows with EHT**, *International Symposium on Computational Sensing 2025*, Hotel Koener, Clervaux, Luxembourg

2024.06.13 **The imaging process of the blackhole shadow with EHT**, *Ricochet days in Lille*, Lille University, France

2024.03.05 **Astronomy and data science**, *International Workshop KEK-Cosmo 2024*, KEK, Tsukuba, Ibaragi, Japan

2024.01.09 **Data science for astronomy**, *Machine Learning at High Energy Physics workshop*, KEK, Tsukuba, Ibaragi, Japan

2023.07.14 **Imaging of ALMA telescope with self-calibration**, *EAS 2034*, Kraków, Poland

2023.06.21 **Data science and imaging the black hole shadows**, *RIKEN AIP seminar*, RIKEN center for AIP, Tokyo, Japan

2023.03.03 **Astronomy and data science: Building astronomical claims with modern data science**, *Interdisciplinary Science Conference in Okinawa (ISCO 2023) — Physics and Mathematics meet Medical Science —*, OIST, Okinawa, Japan

2022.09.13 **Data science and imaging the black hole shadow**, *International online IAU-IAA Astrostats and Astroinfo seminar*, online

2022.02.08 **Toward astro data science**, *Galaxy Evolution Workshop 2021*, NAOJ, Tkyo, Japan

2020.02.20 **Imaging supermassive blackhole with the Event Horizon Telescope**, *The 21th Interdisciplinary Exchange Evening*, RIKEN, Saitama, Japan

2019.10.16 **Data science and astronomy: how it worked for blackhole shadow imaging**, *The cosmos at high energies: exploring extreme physics through novel instrumentation*, Kavli IPMU, Chiba, Japan

2018.09.28 **Optimization theory and astronomy**, *3rd IMI-ISM-ZIB MODAL Workshop on Challenges in Real World Data Analytics and High-Performance Optimization*, GRIPS, Tokyo, Japan

2018.01.11 **Statistical signal processing for astronomy**, *Kavli IPMU-Berkeley Symposium Statistics, Physics and Astronomy*, Kavli IPMU, Chiba, Japan

2017.06.05 **Data scientific approach for astronomy**, *The Third Joint Symposium of Osaka CTSR, RIKEN iTHES/iTHEMS, Kavli IPMU,"Deep Learning and Physics"*, Osaka Univ., Osaka, Japan

2016.06.17 **Cosmological parameter estimation and Fisher information matrix**, *IGAIA IV (Information Geometry and its Applications)*, Liblice, Czech Republic

2016.05.31 **Event classification**, *PhyStat-ν Workshop on Statistical Issues in Experimental Neutrino Physics*, Kavli IPMU, Chiba, Japan

2015.12.15 **Sparse modeling for astronomical data analysis**, *International Meeting on "High-Dimensional Data Driven Science" (HD3-2015)*, Mielparque-Kyoto, Kyoto, Japan

2015.09.25 **Sparsity and information processing**, *MEIS2015 : Mathematical Progress in Expressive Image Synthesis*, Kyushu Univ., Fukuoka, Japan

2014.04.11 **Optimization of probability measures and information geometry**, *2nd International Workshop on Information Geometry and Affine Differential Geometry*, Shanghai, China

2013.08.28 **Optimization of probability measure and its applications in information theory**, *WITMSE2013*, Univ. of Tokyo, Tokyo, Japan

2010.09.23 **Channel capacity problem and optimization of probability measure**, *Mathematical Sciences and Their Applications*, Nagano, Japan

2010.08.03 **Channel capacity and the predictive distribution**, *Information Geometry and its Applications III*, Leipzig, Germany

2009.08.31 **Mean field approximation methods and information geometry**, *Physics of Algorithms '09*, Santa Fe, USA

Grants

2024 – 2029 JSPS Kakenhi, Grant-in-Aid for Specially Promoted Research, 24H00004

Formation processes of heavy elements in the early universe elucidated by superconducting nanoelectronics, large-scale numerical simulations, and data science

Co-Investigator (PI: Kotaro Kohno)

2023 – 2030 JSPS Kakenhi, International Leading Research, 23K20035

Formation processes of heavy elements in the early universe elucidated by superconducting nanoelectronics, large-scale numerical simulations, and data science

Co-Investigator (PI: Kotaro Kohno)

2020 – 2023 JSPS Kakenhi, Grant-in-Aid for Scientific Research (B), 20H01951

New imaging method for ALMA with sparse modeling

Principal Investigator

2020 – 2022 JST, AIP Acceleration Research

Cosmology with big astronomical data using innovative image analysis methods

Co-Investigator (PI: Naoki Yoshida)

2019 – 2024 JSPS Kakenhi, Fund for the Promotion of Joint International Research (B), 19KK0081

Exploring dynamic pictures of supermassive black holes with movie reconstruction from event-horizon-scale observations

Co-Investigator (PI: Mareki Honma)

2017 – 2020 JSPS Kakenhi, Grant-in-Aid for Scientific Research (A), 17H01116

Development of commensal FRB search engine for VERA and observational studies of transient sources

Co-Investigator (PI: Mareki Honma)

2014 – 2019 JST, CREST, Research Area, "Big Data"

Statistical computational cosmology with big astronomical imaging data

Co-Investigator (PI: Naoki Yoshida)

2013 – 2017 MEXT Kakenhi, Grant-in-Aid for Scientific Research on Innovative Areas, "Initiative for high-dimensional data-driven science through deepening of sparse modeling," 25120008

Approach to sparse modeling based on compressed sensing planned

Co-Investigator (PI: Toshiyuki Tanaka)

2013 – 2015 MIC, SCOPE

Research and development of a small-degree-of-freedom interface for multi-degree-of-freedom remote-robot control

Co-Investigator (PI: Jun Morimoto)

2012 – 2014 JSPS Kakenhi, Grant-in-Aid for Scientific Research (C), 24560490

Communication channel capacity and optimization of probability measure

Principal-Investigator

2010 – 2013 JSPS Kakenhi, Grant-in-Aid for Scientific Research (B), 22300098

Mathematical foundation of efficient algorithms for statistical inference

Co-Investigator (PI: Kenji Fukumizu)

2009 – 2012 JSPS Kakenhi, Grant-in-Aid for Scientific Research (B), 21300092

Extraction of synergetic structure of whole body movements and its application to human assisting system

Co-Investigator (PI: Yutaka Sakaguchi)

2007 – 2009 JSPS Kakenhi, Grant-in-Aid for Scientific Research (C), 19500249

Geometric study on statistical learning and computation based on graphs

Co-Investigator (PI: Kenji Fukumizu)

2006 – 2009 MEXT Kakenhi, Grant-in-Aid for Scientific Research on Priority Areas, "Deepening and expansion of statistical mechanical informatics," 18079013

Theoretical analysis and information engineering application of approximate inference methods

Principal-Investigator

2004 – 2006 JSPS Kakenhi, Grant-in-Aid for Young Scientists (B), 16700227

Information geometrical analysis of the cluster variational method

Principal-Investigator

2004 – 2007 JSPS Kakenhi, Grant-in-Aid for Scientific Research (B), 16300088

New statistical methodology for genome diversity analysis

Co-Investigator (PI: Shinto Eguchi)

2003 – 2005 MEXT Kakenhi, Grant-in-Aid for Scientific Research on Priority Areas, "Statistical-mechanical approach to probabilistic information processing," 14084208

Analysis of belief propagation algorithms based on information geometry

Co-Investigator (PI: Yoichi Motomura)

2002 MEXT Kakenhi, Grant-in-Aid for Scientific Research on Priority Areas, "Statistical-mechanical approach to probabilistic information processing," 14084208

Analysis of belief propagation algorithms based on information geometry

Principal-Investigator

1998 – 2001 JST, PRESTO

Mathematical analysis of the EM algorithm and its applications in engineering

Principal-Investigator

Teaching

Cources

2020, 2022, 2024, 2025 **Introduction to Computational Inference**, *Department of Statistical Science, GUAS*, 4 classes, Japanese

2022 **Special Lecture on Radio Astronomy I**, *Astronomical Institute, Tohoku University*, Intensive course, Japanese

2021 **Observational Astronomy, Advanced Course II**, *Department of Astronomy, The University of Tokyo*, Intensive course, Japanese

2020 **Special Lecture on Astronomy IX**, *Department of Astronomical Science, SOKENDAI*, Intensive course, English

2020 **Special Lectures on Mathematical Engineering and Instrumentation Physics**, *Department of Mathematical Engineering and Instrumentation Physics, The University of Tokyo*, 1 class, Japanese

2018 **Interstellar Physics Special Lecture**, *Department of Physics, Nagoya University*, Intensive course, Japanese

2018 **Lecture Series of Data Science in Physics**, *Graduate School of Arts and Sciences, The University of Tokyo*, Intensive course, Japanese

2017 **Lecture Series of Data Science in Astronomy**, *Graduate School of Science, The University of Tokyo*, Intensive course, Japanese

2011, 2012, 2013, 2015 **Advanced Topics in Mathematical Information Sciences II**, *Interdisciplinary Graduate School of Science and Engineering, Tokyo Institute of Technology*, 2 classes, English

2012, 2013 **Compressed sensing: Theory and Applications**, *Department of Informatics, Kyushu University*, Intensive course, Japanese

2005 **Lecture Series of Computational Mathematics 1**, *Graduate School of Information Science, Nagoya University*, Intensive course, Japanese

2001, 2002 **Brain Style Pattern Recognition**, *Department of Brain Science and Engineering, Kyushu Institute of Technology*, Regular course, Japanese

Ph.D Students

Supervise Yuichi Shiraishi (Graduate University of Advanced Studies)

Masatoshi Hamada (Graduate University of Advanced Studies)

Co-supervise Shota Takahashi (Graduate University of Advanced Studies)

Jin Zou (Graduate University of Advanced Studies)

Academic Service

International Conference

2026.03 **Organizing Committee**, *Workshop on Data science for inverse problems and sensing*, ISM, Tachikawa, Tokyo, Japan

2025.03 **Scientific Organizing Committee**, *Study of the Universe by Line Intensity Mapping Experiments (SUBLIME 2025)*, Tokyo, Japan

2025.03 **Organizing Committee**, *Further Developments of Information Geometry 2025*, ISM, The Univ. Tokyo, Tokyo, Japan

2024.10 **Scientific Organizing Committee**, *Data Oriented Astronomy*, The Univ. Tokyo, Tokyo, Japan

2006.03 **International Program Committee**, *6th International Symposium on Independent Component Analysis and Blind Source Separation*, Charleston, USA

2005.12 **Local Organizing Committee**, *2nd International Symposium on Information Geometry and its Applications*, The Univ. Tokyo, Tokyo, Japan

2004.10 **Technical Committee**, *ISCA Tutorial and Research Workshop on Statistical and Perceptual Audio Processing*, Jeju, Korea

2003.04 **International Program Committee**, *4th International Symposium on Independent Component Analysis and Blind Source Separation*, Nara, Japan

Domestic Conference

2023.03 **Organizer**, *Approaches for Inverse Problems*, ISM, Tachikawa, Tokyo

2022.10 **Organizer**, *Data Scientific Methods in Astronomy*, ISM, Tachikawa, Tokyo

2019.05 **Organizer**, *Data Scientific Methods in Astronomy*, ISM, Tachikawa, Tokyo

2017.06 **Organizer**, *Workshop on Minkowski functionals* , Kavli IPMU, Kashiwa, Chiba

2017.05 **Organizer**, *Data Scientific Methods in Astronomy*, ISM, Tachikawa, Tokyo

2015.05 **Organizer**, *Statistical Methods and Astronomy*, ISM, Tachikawa, Tokyo

2014.12 **Organizer**, *Information Geometry for Machine Learning*, RIKEN, Wako, Saitama

2014.03 **Organizer**, *Towards Foundation of Statistical Methods for Medical Image Processing*, ISM Research Collaboration, ISM, Tachikawa, Tokyo

2013.03 **Organizer**, *Optimization of Probability Measure and Communication Channel Capacity*, ISM Research Collaboration, ISM, Tachikawa, Tokyo

Editorial Board

2025 **Special Issue Managing Editor**, *Information Geometry*, Springer, Special issue on "2025 Further Developments of Information Geometry"

2018 – **Associate editor**, *Information Geometry*, Springer

2008 – 2017 **Associate editor**, *Annals of the Institute of Statistical Mathematics*, Springer

2007 **Editor of featured section**, *Special Issue: Information Geometry and Its Applications, Annals of the Institute of Statistical Mathematics, Vol.59. No.1*, Springer

2006 – 2013 **Editor**, *Neural Networks*, Elsevier

2005 – 2014 **Associate editor**, *IEEE transactions on Neural Networks and Learning Systems*

2005 – 2008 **Co-editor**, *Annals of the Institute of Statistical Mathematics*, Springer

Journal Reviews

IEEE transactions (Biological Engineering, Communications, Information Theory, Neural Networks, and Signal and Audio Processing), IEEE Proceedings, Neurocomputing, IEICE transactions, Signal Processing, Neural Networks, Physical Review Research, Publications of the Astronomical Society of Japan, Astronomy & Astrophysics, The Astronomical Journal, Journal of the Royal Statistical Society B, Monthly Notices of the Royal Astronomical Society, Information geometry, Entropy, Journal of Machine Learning Research, Nature Methods

Conference Reviews

Artificial Intelligence and Statistics, Neural Information Processing Systems (NIPS) (1999–2015), International Conference on Machine Learning (ICML) (2007), A series of ICA (Independent Component Analysis and Blind Source Separation) conferences, IEEE International Symposium on Information Theory (ISIT), IEEE Information Theory Workshop (ITW), IEEE International Joint Conference on Neural Networks (IJCNN), International Conference on Artificial Neural Networks, International Conference on Neural Information Processing (ICONIP 2025)

Administrative Committee

2019 – 2022 **Committee Member**, *Science Advisory Committee*, National Astronomical Observatory of Japan

2015 – 2016 **Examination Committee Member**, *Special Research Fellowship for Young Scientists*, Japan Society for Promotion of Science

2014 – 2015 **Committee Member**, *Technical Committee on Information Theory*, The Institute of Electronics, Information and Communication Engineering

Publications

Articles in journals

[1] V. Bernshteyn, N. S. Conroy, M. Bauböck, P. Tiede, A. V. Joshi, B. S. Prather, C. F. Gammie, and The Event Horizon Telescope Collaboration. "Ring Asymmetry and Spin in M87*". In: *The Astrophysical Journal* 1000.2 (Mar. 2026). (7 leading authors and 257 EHT collaboration members including S.I.), 231(21pp). DOI: 10.3847/1538-4357/ae34af. arXiv: 2601.00394 [astro-ph.HE].

[2] Saurabh, H. Müller, S. D. von Fellenberg, P. Tiede, M. Janssen, L. Blackburn, A. E. Broderick, et al. "Probing jet base emission of M87* with the 2021 Event Horizon Telescope observations". In: *Astronomy & Astrophysics* 706 (Jan. 2026). (17 leading authors and 269 EHT collaboration members including S.I.), A27(20pp). DOI: 10.1051/0004-6361/202557022. arXiv: 2512.08970 [astro-ph.HE].

[3] S. Takahashi, M. Tanaka, and S. Ikeda. "Majorization-minimization Bregman proximal gradient algorithms for NMF with the Kullback–Leibler divergence". In: *Journal of Optimization Theory and Applications* 208 (Jan. 2026), 14(34pp). DOI: 10.1007/s10957-025-02833-y. arXiv: 2405.11185 [math.OC].

[4] Gómez, José L., Cho, Ilje, Traianou, Efthalia, Krichbaum, Thomas P., Zhao, Guang-Yao, Lico, Rocco, Fuentes, Antonio, et al. "Spatially resolved polarization swings in the supermassive binary black hole candidate OJ 287 with first Event Horizon Telescope observations". In: *Astronomy & Astrophysics* 705 (Jan. 2026). (26 leading authors and 257 EHT collaboration members including S.I.), A23(24pp). DOI: 10.1051/0004-6361/202555831.

[5] The Event Horizon Telescope Collaboration. "Horizon-scale variability of M87* from 2017–2021 EHT observations". In: *Astronomy & Astrophysics* 704 (Dec. 2025). (304 authors in alphabetical order), A91(37pp). DOI: 10.1051/0004-6361/202555855.

[6] Y. Tsuzuki, S. Ikeda, H. Yoneda, and T. Takahashi. "Langevin Compton Imaging: A new method of visualizing radioactive sources based on Markov chain Monte Carlo". In: *Nuclear Instruments and Methods in Physics Research A* 1080 (Nov. 2025), 170815(9pp). DOI: 10.1016/j.nima.2025.170815.

[7] J. P. Pedroso and S. Ikeda. "Maximum-expectation matching under recourse". In: *European Journal of Operational Research* 324.3 (Aug. 2025), pp. 732–741. DOI: 10.1016/j.ejor.2025.02.012. arXiv: 1605.08616 [cs.DS].

[8] C. Goddi, D. F. Carlos, G. B. Crew, L. D. Matthews, H. Messias, A. Mus, I. Martí-Vidal, et al. "First polarization study of the M87 jet and active galactic nuclei at submillimeter wavelengths with ALMA". In: *Astronomy & Astrophysics* 699 (July 2025). (12 leading authors and 240 EHT collaboration members including S.I.), A265(17pp). DOI: 10.1051/0004-6361/202554140. arXiv: 2505.10181 [astro-ph.GA].

[9] R. Dahale, I. Cho, K. Moriyama, K. Wiik, P. Tiede, J. L. Gómez, C.-K. Chan, et al. "Origin of the Ring Ellipticity in the Black Hole Images of M87*". In: *Astronomy & Astrophysics* 699 (July 2025). (19 leading authors and 257 EHT collaboration members including S.I.), A279(18pp). DOI: 10.1051/0004-6361/202555235. arXiv: 2505.10333 [astro-ph.HE].

[10] J. Röder, M. Wielgus, A. P. Lobanov, T. P. Krichbaum, D. G. Nair, S.-S. Lee, E. Ros, et al. "A multifrequency study of sub-parsec jets with the Event Horizon Telescope". In: *Astronomy & Astrophysics* 695 (Mar. 2025). (31 leading authors and 259 EHT collaboration members including S.I.), A233(21pp). DOI: 10.1051/0004-6361/202452600. arXiv: 2501.05518 [astro-ph.HE].

[11] M. Uemura, Y. Koga, R. Sazaki, T. Yukino, T. Nakaoka, R. Imazawa, T. Kato, et al. "Smart Kanata: A Framework for Autonomous Decision Making in Rapid Follow-up Observations of Cataclysmic Variables". In: *Publications of the Astronomical Society of Japan* 77.2 (Feb. 2025). 14th/14 authors, pp. 219–236. DOI: 10.1093/pasj/psae112. arXiv: 2412.02092 [astro-ph.SR].

[12] S. Ikeda, T. Nakazato, T. Tsukagoshi, T. T. Takeuchi, and M. Yamaguchi. "Solving Self-calibration of ALMA Data with an Optimization Method". In: *Publications of the Astronomical Society of Japan* 77.2 (Feb. 2025), pp. 260–276. DOI: 10.1093/pasj/psae114. arXiv: 2412.03183 [astro-ph.IM].

[13] The Event Horizon Telescope Collaboration. "The Persistent Shadow of the Supermassive Black Hole of M87, II. Model comparisons and theoretical interpretations". In: *Astronomy & Astrophysics* 681 (Jan. 2025). (269 authors in alphabetical order), A265(27pp). DOI: 10.1051/0004-6361/202451296.

[14] J. C. Algaba, M. Baloković, S. Chandra, W.-Y. Cheong, Y.-Z. Cui, F. D'Ammando, A. D. Falcone, et al. "Broadband multi-wavelength properties of M87 during the 2018 EHT campaign including a very high energy flaring episode". In: *Astronomy & Astrophysics* 692 (Dec. 2024). (32 leading authors and 729 authors including S.I.), A140(44pp). DOI: 10.1051/0004-6361/202452600. arXiv: 2404.17623 [astro-ph.HE].

[15] The Event Horizon Telescope Collaboration. "First Sagittarius A* Event Horizon Telescope Results. VII. Polarization of the Ring". In: *The Astrophysical Journal Letters* 964.2 (Mar. 2024). (285 authors in alphabetical order), L25(32pp). ISSN: 2041-8205. DOI: 10.3847/2041-8213/ad2df0.

[16] The Event Horizon Telescope Collaboration. "First Sagittarius A* Event Horizon Telescope Results. VIII. Physical Interpretation of the Polarized Ring". In: *The Astrophysical Journal Letters* 964.2 (Mar. 2024). (286 authors in alphabetical order), L26(37pp). ISSN: 2041-8205. DOI: 10.3847/2041-8213/ad2df1.

[17] M. Shirasaki and S. Ikeda. "Neural style transfer of weak lensing mass maps". In: *The Open Journal of Astrophysics* 7 (May 2024). DOI: 10.33232/001c.118104. arXiv: 2310.17141 [astro-ph.CO].

[18] M. Yamaguchi, T. Muto, T. Tsukagoshi, H. Nomura, N. Hirano, T. Nakazato, S. Ikeda, et al. "ALMA 2D Super-resolution Imaging of Taurus-Auriga Protoplanetary Disks: Probing Statistical Properties of Disk Substructures". In: *Publications of the Astronomical Society of Japan* 76.3 (May 2024). (7th/9 authors), pp. 437–474. DOI: 10.1093/pasj/psae022. arXiv: 2404.13570 [astro-ph.EP].

[19] A. W. Raymond, S. S. Doeleman, K. Asada, L. Blackburn, G. C. Bower, M. Bremer, D. Broguiere, et al. "First Very Long Baseline Interferometry Detections at 870 μm". In: *The Astronomical Journal* 168.3 (Aug. 2024). (39 leading authors and 240 EHT collaboration members including S.I.), 130(19pp). DOI: 10.3847/1538-3881/ad5bdb.

[20] G. F. Paraschos, J.-Y. Kim, M. Wielgus, J. Röder, T. P. Krichbaum, E. Ros, I. Agudo, et al. "Ordered magnetic fields around the 3C 84 central black hole". In: *Astronomy & Astrophysics* 682 (Feb. 2024). (17 leading authors and 266 EHT collaboration members including S.I.), L3(15pp). DOI: 10.1051/0004-6361/202348308. arXiv: 2402.00927 [astro-ph.HE].

[21] The Event Horizon Telescope Collaboration. "The Persistent Shadow of the Supermassive Black Hole of M87, I. Observations, calibration, imaging, and analysis". In: *Astronomy & Astrophysics* 681 (Jan. 2024). (309 authors in alphabetical order), A79(63pp). DOI: 10.1051/0004-6361/202347932.

[22] P. Torne, K. Liu, R. P. Eatough, J. Wongphechauxsorn, J. M. Cordes, G. Desvignes, M. De Laurentis, et al. "A Search for Pulsars around Sgr A* in the First Event Horizon Telescope Data Set". In: *The Astrophysical Journal* 959.1 (Nov. 2023). (42 leading authors and 243 EHT collaboration members including S.I.), 14(27pp). DOI: 10.3847/1538-4357/acf4f2. arXiv: 2308.15381 [astro-ph.HE].

[23] F. Roelofs, M. D. Johnson, A. Chael, M. Janssen, M. Wielgus, A. E. Broderick, and The Event Horizon Telescope Collaboration. "Polarimetric Geometric Modeling for mm-VLBI Observations of Black Holes". In: *The Astrophysical Journal Letters* 957.2 (Nov. 2023). (6 leading authors and 277 EHT collaboration members including S.I.), L21(29pp). ISSN: 2041-8205. DOI: 10.3847/2041-8213/acff6f. arXiv: 2311.10695 [astro-ph.HE].

[24] The Event Horizon Telescope Collaboration. "First M87 Event Horizon Telescope Results. IX. Detection of Near-horizon Circular Polarization". In: *The Astrophysical Journal Letters* 957.2 (Nov. 2023). (283 authors in alphabetical order), L20(42pp). ISSN: 2041-8205. DOI: 10.3847/2041-8213/acff70. arXiv: 2205.06460 [astro-ph.HE].

[25] B. S. Prather, J. Dexter, M. Moscibrodzka, H.-Y. Pu, T. Bronzwaer, J. Davelaar, Z. Younsi, et al. "Comparison of Polarized Radiative Transfer Codes Used by the EHT Collaboration". In: *The Astrophysical Journal* 950.1 (June 2023). (10 leading authors and 263 EHT collaboration members including S.I.), 35(23pp). DOI: 10.3847/1538-4357/acc586.

[26] S. Jorstad, M. Wielgus, R. Lico, S. Issaoun, A. E. Broderick, D. W. Pesce, J. Liu, et al. "The Event Horizon Telescope Image of the Quasar NRAO 530". In: *The Astrophysical Journal* 943.2 (Feb. 2023). (29 leading authors and 241 EHT collaboration members including S.I.), 170(27pp). DOI: 10.3847/1538-4357/acaea8. arXiv: 2302.04622 [astro-ph.HE].

[27] S. Takahashi, M. Tanaka, and S. Ikeda. "Blind Deconvolution with Non-smooth Regularization via Bregman Proximal DCAs". In: *Signal Processing* 202 (Jan. 2023), 108734(9pp). DOI: 10.1016/j.sigpro.2022.108734. arXiv: 2205.06460 [math.OC].

[28] S. Issaoun, M. Wielgus, S. Jorstad, T. P. Krichbaum, L. Blackburn, M. Janssen, C.-K. Chan, et al. "Resolving the Inner Parsec of the Blazar J1924–2914 with the Event Horizon Telescope". In: *The Astrophysical Journal* 934.2 (Aug. 2022). (29 leading authors and 239 EHT collaboration members including S.I.), 145(21pp). DOI: 10.3847/1538-4357/ac7a40.

[29] The Event Horizon Telescope Collaboration. "First Sagittarius A* Event Horizon Telescope Results. I. The Shadow of the Supermassive Black Hole in the Center of the Milky Way". In: *The Astrophysical Journal Letters* 930.2 (May 2022). (388 authors in alphabetical order), L12(21pp). ISSN: 2041-8205. DOI: 10.3847/2041-8213/ac6674. arXiv: 2311.08680.

[30] The Event Horizon Telescope Collaboration. "First Sagittarius A* Event Horizon Telescope Results. II. EHT and Multiwavelength Observations, Data Processing, and Calibration". In: *The Astrophysical Journal Letters* 930.2 (May 2022). (336 authors in alphabetical order), L13(31pp). ISSN: 2041-8205. DOI: 10.3847/2041-8213/ac6675. arXiv: 2311.08679.

[31] The Event Horizon Telescope Collaboration. "First Sagittarius A* Event Horizon Telescope Results. III. Imaging of the Galactic Center Supermassive Black Hole". In: *The Astrophysical Journal Letters* 930.2 (May 2022). (270 authors in alphabetical order), L14(64pp). ISSN: 2041-8205. DOI: 10.3847/2041-8213/ac6429. arXiv: 2311.09479.

[32] The Event Horizon Telescope Collaboration. "First Sagittarius A* Event Horizon Telescope Results. IV. Variability, Morphology, and Black Hole Mass". In: *The Astrophysical Journal Letters* 930.2 (May 2022). (269 authors in alphabetical order), L15(52pp). ISSN: 2041-8205. DOI: 10.3847/2041-8213/ac6736. arXiv: 2311.08697.

[33] The Event Horizon Telescope Collaboration. "First Sagittarius A* Event Horizon Telescope Results. V. Testing Astrophysical Models of the Galactic Center Black Hole". In: *The Astrophysical Journal Letters* 930.2 (May 2022). (274 authors in alphabetical order), L16(49pp). ISSN: 2041-8205. DOI: 10.3847/2041-8213/ac6672. arXiv: 2311.09478.

[34] The Event Horizon Telescope Collaboration. "First Sagittarius A* Event Horizon Telescope Results. VI. Testing the Black Hole Metric". In: *The Astrophysical Journal Letters* 930.2 (May 2022). (270 authors in alphabetical order), L17(44pp). ISSN: 2041-8205. DOI: 10.3847/2041-8213/ac6756. arXiv: 2311.09484.

[35] J. Farah, P. Galison, K. Akiyama, K. L. Bouman, G. C. Bower, A. Chael, A. Fuentes, et al. "Selective Dynamical Imaging of Interferometric Data". In: *The Astrophysical Journal Letters* 930.2 (May 2022). (18 leading authors and 216 EHT collaboration members including S.I.), L18(21pp). ISSN: 2041-8205. DOI: 10.3847/2041-8213/ac6615.

[36] M. Wielgus, N. Marchili, I. Martí-Vidal, G. K. Keating, V. Ramakrishnan, P. Tiede, E. Fomalont, et al. "Millimeter Light Curves of Sagittarius A* Observed during the 2017 Event Horizon Telescope Campaign". In: *The Astrophysical Journal Letters* 930.2 (May 2022). (31 leading authors and 236 EHT collaboration members include S.I.), L19(32pp). ISSN: 2041-8205. DOI: 10.3847/2041-8213/ac6428.

[37] B. Georgiev, D. W. Pesce, A. E. Broderick, G. N. Wong, V. Dhruv, M. Wielgus, C. F. Gammie, et al. "A Universal Power-law Prescription for Variability from Synthetic Images of Black Hole Accretion Flows". In: *The Astrophysical Journal Letters* 930.2 (May 2022). (30 leading authors and 239 EHT collaboration members including S.I.), L20(32pp). ISSN: 2041-8205. DOI: 10.3847/2041-8213/ac65eb.

[38] A. E. Broderick, R. Gold, B. Georgiev, D. W. Pesce, P. Tiede, C. Ni, K. Moriyama, and The Event Horizon Telescope Collaboration. "Characterizing and Mitigating Intraday Variability: Reconstructing Source Structure in Accreting Black Holes with mm-VLBI". In: *The Astrophysical Journal Letters* 930.2 (May 2022). (7 leading authors and 260 EHT collaboration members including S.I.), L21(30pp). ISSN: 2041-8205. DOI: 10.3847/2041-8213/ac6584.

[39] K. Satapathy, D. Psaltis, F. Özel, L. Medeiros, S. T. Dougall, C.-K. Chan, M. Wielgus, et al. "The Variability of the Black Hole Image in M87 at the Dynamical Timescale". In: *The Astrophysical Journal* 925.1 (Jan. 2022). (10 leading authors and 228 EHT collaboration members including S.I.), 13(19pp). DOI: 10.3847/1538-4357/ac332e. arXiv: 2111.01317 [astro-ph.HE].

[40] M. Yamaguchi, T. Tsukagoshi, T. Muto, H. Nomura, T. Nakazato, S. Ikeda, M. Tamura, and R. Kawabe. "ALMA Super-resolution Imaging of T Tau: $r = 12$ au Gap in the Compact Dust Disk around T Tau N". In: *The Astrophysical Journal* 923.1 (Dec. 2021), p. 121. DOI: 10.3847/1538-4357/ac2bfd. arXiv: 2110.00974 [astro-ph.EP].

[41] A. Taniguchi, Y. Tamura, S. Ikeda, T. Takekoshi, and R. Kawabe. "A Data-scientific Noise-removal Method for Efficient Submillimeter Spectroscopy With Single-dish Telescopes". In: *The Astronomical Journal* 162.3 (Aug. 2021), p. 111. DOI: 10.3847/1538-3881/ac11f7. arXiv: 2107.06290 [astro-ph.IM].

[42] X. Li, N. Yoshida, M. Oguri, S. Ikeda, and W. Luo. "Three-dimensional Reconstruction of Weak-lensing Mass Maps with a Sparsity Prior. I. Cluster Detection". In: *The Astrophysical Journal* 916.2 (July 2021), 67(13pp). DOI: 10.3847/1538-4357/ac0625. arXiv: 2102.09707 [astro-ph.CO].

[43] M. Janssen, H. Falcke, M. Kadler, E. Ros, M. Wielgus, K. Akiyama, M. Baloković, et al. "Event Horizon Telescope observations of the jet launching and collimation in Centaurus A". In: *Nature Astronomy* 5 (July 2021). (35 leading authors and 236 EHT collaboration members including S.I.), pp. 1017–1028. DOI: 10.1038/s41550-021-01417-w.

[44] T. Omama, M. Uemura, S. Ikeda, and M. Morii. "Extracting common signal components from the X-ray and optical light curves of GX 339-4: new view for anti-correlation". In: *Publications of the Astronomical Society of Japan* 73.3 (June 2021), pp. 716–727. DOI: 10.1093/pasj/psab032. arXiv: 2104.07338 [astro-ph.HE].

[45] P. Kocherlakota, L. Rezzolla, H. Falcke, C. M. Fromm, M. Kramer, Y. Mizuno, A. Nathanail, et al. "Constraints on black-hole charges with the 2017 EHT observations of M87*". In: *Physical Review D* 103.10 (May 2021). (9 leading authors and 227 EHT collaboration members including S.I.), 104047(18pp). DOI: 10.1103/PhysRevD.103.104047.

[46] M. Shirasaki, K. Moriwaki, T. Oogi, N. Yoshida, S. Ikeda, and T. Nishimichi. "Noise reduction for weak lensing mass mapping: An application of generative adversarial networks to Subaru Hyper Suprime-Cam first-year data". In: *Monthly Notices of the Royal Astronomical Society* (Apr. 2021), stab982. DOI: 10.1093/mnras/stab982. arXiv: 1911.12890 [astro-ph.CO].

[47] The Event Horizon Telescope Collaboration. "First M87 Event Horizon Telescope Results. VIII. Magnetic Field Structure near The Event Horizon". In: *The Astrophysical Journal Letters* 910.1 (Mar. 2021). (240 authors in alphabetical order), L13(43pp). ISSN: 2041-8205. DOI: 10.3847/2041-8213/abe4de. arXiv: 2105.01173 [astro-ph.HE].

[48] The Event Horizon Telescope Collaboration. "First M87 Event Horizon Telescope Results. VII. Polarization of the Ring". In: *The Astrophysical Journal Letters* 910.1 (Mar. 2021). (239 authors in alphabetical order), L12(48pp). ISSN: 2041-8205. DOI: 10.3847/2041-8213/abe71d. arXiv: 2105.01169 [astro-ph.HE].

[49] R. Ohsawa, A. Hirota, K. Morita, S. Abe, D. Kastinen, J. Kero, C. Szasz, et al. "Relationship between radar cross section and optical magnitude based on radar and optical simultaneous observations of faint meteors". In: *Planetary and Space Science* 194 (Dec. 2020). (18th/43 authors), p. 105011. ISSN: 0032-0633. DOI: 10.1016/j.pss.2020.105011. arXiv: 2008.08942 [astro-ph.EP].

[50] M. Aizawa, Y. Suto, Y. Oya, S. Ikeda, and T. Nakazato. "Search for Alignment of Disk Orientations in Nearby Star-forming Regions: Lupus, Taurus, Upper Scorpius, ρ Ophiuchi, and Orion". In: *The Astrophysical Journal* 899.1 (Aug. 2020), p. 55. DOI: 10.3847/1538-4357/aba43d. arXiv: 2007.03393 [astro-ph.SR].

[51] J.-Y. Kim, T. P. Krichbaum, A. Broderick, M. Wielgus, L. Blackburn, J. L. Gómez, M. D. Johnson, et al. "Event Horizon Telescope imaging of the archetypal blazar 3C 279 at an extreme 20 microarcsecond resolution". In: *Astronomy & Astrophysics* 640 (Aug. 2020). (18 leading authors and 334 EHT collaboration members including S.I.), A69(21pp). DOI: 10.1051/0004-6361/202037493.

[52] M. Yamaguchi, K. Akiyama, T. Tsukagoshi, T. Muto, A. Kataoka, F. Tazaki, S. Ikeda, et al. "Super-resolution Imaging of the Protoplanetary Disk HD 142527 Using Sparse Modeling". In: *The Astrophysical Journal* 895.2 (May 2020). (7th/10 authors), 84(14pp). DOI: 10.3847/1538-4357/ab899f. arXiv: 2004.10989 [astro-ph.EP].

[53] M. Uemura, T. Abe, Y. Yamada, and S. Ikeda. "Feature selection for classification of blazars based on optical photometric and polarimetric time-series data". In: *Publications of the Astronomical Society of Japan* 72.5 (Oct. 2020), p. 74. DOI: 10.1093/pasj/psaa063. arXiv: 2006.07792 [astro-ph.HE].

[54] Roelofs, F., Janssen, M., Natarajan, I., Deane, R., Davelaar, J., Olivares, H., Porth, O., et al. "SYMBA: An end-to-end VLBI synthetic data generation pipeline - Simulating Event Horizon Telescope observations of M 87". In: *Astronomy & Astrophysics* 636 (Apr. 2020). (12 leading authords and 196 EHT collboration members including S.I.), A5(19pp). DOI: 10.1051/0004-6361/201936622.

[55] M. W. Richmond, M. Tanaka, T. Morokuma, S. Sako, R. Ohsawa, N. Arima, N. Tominaga, et al. "An optical search for transients lasting a few seconds". In: *Publications of the Astronomical Society of Japan* 72.1 (Feb. 2020). (12th/40 authors), p. 3. DOI: 10.1093/pasj/psz120. arXiv: 1910.11343 [astro-ph.IM].

[56] T. Kasai, S. Ono, S. Koshiba, M. Yamamoto, T. Tanaka, S. Ikeda, and T. Kigawa. "Amino-acid selective isotope labeling enables simultaneous overlapping signal decomposition and information extraction from NMR spectra". In: *Journal of Biomolecular NMR* 74 (Jan. 2020), pp. 125–137. ISSN: 1573-5001. DOI: 10.1007/s10858-019-00295-9.

[57] Y. Maeda, R. Iizuka, T. Hayashi, T. Sato, N. Nakaniwa, M. Takeo, H. Suzuki, et al. "A concept for X-ray telescope system with an angular-resolution booster". In: *Publications of the Astronomical Society of Japan* 71.5 (Oct. 2019). (9th/10 authors), p. 97. DOI: 10.1093/pasj/psz081. arXiv: 1908.00231 [astro-ph.IM].

[58] M. Shirasaki, N. Yoshida, and S. Ikeda. "Denoising Weak Lensing Mass Maps with Deep Learning". In: *Physical Review D* 100 (Aug. 2019), p. 043527. DOI: 10.1103/PhysRevD.100.043527. arXiv: 1812.05781 [astro-ph.CO].

[59] O. Porth, K. Chatterjee, R. Narayan, C. F. Gammie, Y. Mizuno, P. Anninos, J. G. Baker, et al. "The Event Horizon General Relativistic Magnetohydrodynamic Code Comparison Project". In: *The Astronomical Journal Supplement Series* 243.2 (Aug. 2019). (27 leading authors and 192 EHT collaboration members including S.I.), 26(40pp). ISSN: 0067-0049. DOI: 10.3847/1538-4365/ab29fd. arXiv: 1904.04923 [astro-ph.HE].

[60] N. Yasuda, M. Tanaka, N. Tominaga, J. Jiang, T. J. Moriya, T. Morokuma, N. Suzuki, et al. "The Hyper Suprime-Cam SSP Transient Survey in COSMOS: Overview". In: *Publications of the Astronomical Society of Japan* 71.4 (Aug. 2019). (12th/21 authors), p. 74. DOI: 10.1093/pasj/psz050. arXiv: 1904.09697 [astro-ph.GA].

[61] K. Arimatsu, R. Ohsawa, G. L. Hashimoto, S. Urakawa, J. Takahashi, M. Tozuka, Y. Itoh, et al. "New Constraint on the Atmosphere of (50000) Quaoar from a Stellar Occultation". In: *The Astronomical Journal* 158.6 (Nov. 2019). (14th/38 authors), p. 236. DOI: 10.3847/1538-3881/ab5058. arXiv: 1910.09988 [astro-ph.EP].

[62] The Event Horizon Telescope Collaboration. "First M87 Event Horizon Telescope Results. I. The Shadow of the Supermassive Black Hole". In: *The Astrophysical Journal Letters* 875.1 (Apr. 2019). (348 authors in alphabetical order), L1(17pp). ISSN: 2041-8205. DOI: 10.3847/2041-8213/ab0ec7. arXiv: 1906.11238 [astro-ph.GA].

[63] The Event Horizon Telescope Collaboration. "First M87 Event Horizon Telescope Results. II. Array and Instrumentation". In: *The Astrophysical Journal Letters* 875.1 (Apr. 2019). (341 authors in alphabetical order), L2(28pp). ISSN: 2041-8205. DOI: 10.3847/2041-8213/ab0c96. arXiv: 1906.11239 [astro-ph.IM].

[64] The Event Horizon Telescope Collaboration. "First M87 Event Horizon Telescope Results. III. Data Processing and Calibration". In: *The Astrophysical Journal Letters* 875.1 (Apr. 2019). (217 authors in alphabetical order), L3(32pp). ISSN: 2041-8205. DOI: 10.3847/2041-8213/ab0c57. arXiv: 1906.11240 [astro-ph.GA].

[65] The Event Horizon Telescope Collaboration. "First M87 Event Horizon Telescope Results. IV. Imaging the Central Supermassive Black Hole". In: *The Astrophysical Journal Letters* 875.1 (Apr. 2019). (215 authors in alphabetical order), L4(52pp). ISSN: 2041-8205. DOI: 10.3847/2041-8213/ab0e85. arXiv: 1906.11241 [astro-ph.GA].

[66] The Event Horizon Telescope Collaboration. "First M87 Event Horizon Telescope Results. V. Physical Origin of the Asymmetric Ring". In: *The Astrophysical Journal Letters* 875.1 (Apr. 2019). (220 authors in alphabetical order), L5(31pp). ISSN: 2041-8205. DOI: 10.3847/2041-8213/ab0f43. arXiv: 1906.11242 [astro-ph.GA].

[67] The Event Horizon Telescope Collaboration. "First M87 Event Horizon Telescope Results. VI. The Shadow and Mass of the Central Black Hole". In: *The Astrophysical Journal Letters* 875.1 (Apr. 2019). (214 authors in alphabetical order), L6(44pp). ISSN: 2041-8205. DOI: 10.3847/2041-8213/ab1141. arXiv: 1906.11243 [astro-ph.GA].

[68] R. Ohsawa, S. Sako, Y. Sarugaku, F. Usui, T. Ootsubo, Y. Fujiwara, M. Sato, et al. "Luminosity function of faint sporadic meteors measured with a wide-field CMOS mosaic camera Tomo-e PM". In: *Planetary and Space Science* 165 (Jan. 2019). (33rd/37 authors), pp. 281–292. ISSN: 0032-0633. DOI: 10.1016/j.pss.2018.09.006. arXiv: 1809.08816 [astro-ph].

[69] M. Morii, S. Ikeda, and Y. Maeda. "An image reconstruction method for X-ray telescope system with an angular resolution booster". In: *Publications of the Astronomical Society of Japan* 71.1 (Jan. 2019), p. 24. DOI: 10.1093/pasj/psy143. arXiv: 1811.08653 [astro-ph.IM].

[70] K. Kuramochi, K. Akiyama, S. Ikeda, F. Tazaki, V. L. Fish, H.-Y. Pu, K. Asada, and M. Honma. "Superresolution Interferometric Imaging with Sparse Modeling Using Total Squared Variation – Application to Imaging the Black Hole Shadow". In: *The Astrophysical Journal* 858.1 (May 2018), p. 56. DOI: 10.3847/1538-4357/aab6b5. arXiv: 1802.05783 [astro-ph.IM].

[71] R. Ohsawa, S. Sako, T. Miyata, T. Kamizuka, K. Okada, K. Mori, M. S. Uchiyama, et al. ""Slow-scanning" in ground-based mid-infrared observations". In: *The Astrophysical Journal* 857.1 (Apr. 2018). (11th/11 authors), 37(11pp). DOI: 10.3847/1538-4357/aab6ae. arXiv: 1804.04271 [astro-ph].

[72] Y. Igarashi, H. Takenaka, Y. Nakanishi-Ohno, M. Uemura, S. Ikeda, and M. Okada. "Exhaustive search for sparse variable selection in linear regression". In: *Journal of the Physical Society of Japan* 87.4 (Mar. 2018), p. 044802. DOI: 10.7566/JPSJ.87.044802. arXiv: 1707.02050 [stat.ML].

[73] T. Obuchi, S. Ikeda, K. Akiyama, and Y. Kabashima. "Accelerating cross-validation with total variation and its application to super-resolution imaging". In: *PLOS ONE* 12.12 (Dec. 2017). e0188012, pp. 1–14. DOI: 10.1371/journal.pone.0188012. arXiv: 1611.07197 [stat.ME].

[74] K. Akiyama, K. Kuramochi, S. Ikeda, V. L. Fish, F. Tazaki, M. Honma, S. S. Doeleman, et al. "Imaging the Schwarzschild-radius-scale Structure of M87 with the Event Horizon Telescope using Sparse Modeling". In: *The Astrophysical Journal* 838.1 (Mar. 2017). (3rd/13 authors), 1(16pp). DOI: 10.3847/1538-4357/aa6305. arXiv: 1702.07361 [astro-ph.IM].

[75] K. Akiyama, S. Ikeda, M. Pleau, V. L. Fish, F. Tazaki, K. Kuramochi, A. E. Broderick, et al. "Superresolution Full-polarimetric Imaging for Radio Interferometry with Sparse Modeling". In: *The Astronomical Journal* 153.4 (Mar. 2017). (2nd/12 authors), 159(12pp). DOI: 10.3847/1538-3881/aa6302. arXiv: 1702.00424 [astro-ph.IM].

[76] M. Morii, S. Ikeda, S. Sako, and R. Ohsawa. "Data Compression for the Tomo-e Gozen Using Low-rank Matrix Approximation". In: *The Astrophysical Journal* 835.1 (Jan. 2017), 1(5pp). DOI: 10.3847/1538-4357/835/1/1. arXiv: 1612.03994 [astro-ph.IM].

[77] T. Ikeya, T. Hanashima, S. Hosoya, M. Shimazaki, S. Ikeda, M. Mishima, P. Güntert, and Y. Ito. "Improved in-cell structure determination of proteins at near-physiological concentration". In: *Scientific Reports* 6 (Dec. 2016), 38312(11pp). DOI: 10.1038/srep38312.

[78] T. Ikeya, S. Ikeda, T. Kigawa, Y. Ito, and P. Güntert. "Protein NMR Structure Refinement based on Bayesian Inference". In: *Journal of Physics: Conference Series* 699 (Dec. 2016). International Meeting on High-Dimensional Data-Driven Science (HD3-2015) 14-17 Dec. 2015, Kyoto, Japan, 012005(14pp). DOI: 10.1088/1742-6596/699/1/012005.

[79] M. Honma, K. Akiyama, F. Tazaki, K. Kuramochi, S. Ikeda, K. Hada, and M. Uemura. "Imaging black holes with sparse modeling". In: *Journal of Physics: Conference Series* 699 (Dec. 2016). International Meeting on High-Dimensional Data-Driven Science (HD3-2015) 14-17 Dec. 2015, Kyoto, Japan, 012006(9pp). DOI: 10.1088/1742-6596/699/1/012006.

[80] S. Ikeda, H. Odaka, and M. Uemura. "Sparse Modeling for Astronomical Data Analysis". In: *Journal of Physics: Conference Series* 699 (Dec. 2016). International Meeting on High-Dimensional Data-Driven Science (HD3-2015) 14-17 Dec. 2015, Kyoto, Japan, 012008(4pp). DOI: 10.1088/1742-6596/699/1/012008.

[81] M. Uemura, K. S. Kawabata, S. Ikeda, K. Maeda, H.-Y. Wu, K. Watanabe, S. Takahashi, and I. Fujishiro. "Data-driven approach to Type Ia supernovae: variable selection on the peak luminosity and clustering in visual analytics". In: *Journal of Physics: Conference Series* 699 (Dec. 2016). International Meeting on High-Dimensional Data-Driven Science (HD3-2015) 14-17 Dec. 2015, Kyoto, Japan, 012009(10pp). DOI: 10.1088/1742-6596/699/1/012009.

[82] K. Watanabe and S. Ikeda. "Rate-Distortion Functions for Gamma-type Sources under Absolute-log Distortion Measure". In: *IEEE Transactions on Information Theory* 62.10 (Oct. 2016), pp. 5496–5502. DOI: 10.1109/TIT.2016.2602100.

[83] M. Morii, S. Ikeda, N. Tominaga, M. Tanaka, T. Morokuma, K. Ishiguro, J. Yamato, et al. "Machine-learning Selection of Optical Transients in Subaru/Hyper Suprime-Cam Survey". In: *Publications of the Astronomical Society of Japan* 68.6 (Oct. 2016). (2nd/11 authors), 104(8pp). DOI: 10.1093/pasj/psw096. arXiv: 1609.03249 [astro-ph.IM].

[84] S. Ikeda, F. Tazaki, K. Akiyama, K. Hada, and M. Honma. "PRECL: A new method for interferometry imaging from closure phase". In: *Publications of the Astronomical Society of Japan* 68.3 (Apr. 2016), 45(9pp). DOI: 10.1093/pasj/psw042. arXiv: 1603.07078 [astro-ph.IM].

[85] H. Okamura, S. Ikeda, T. Morita, and S. Eguchi. "Risk assessment of radioisotope contamination for aquatic living resources in and around Japan". In: *Proceedings of the National Academy of Sciences (PNAS)* 113.14 (Feb. 2016), pp. 3838–3843. DOI: 10.1073/pnas.1519792113.

[86] O. Komori, S. Eguchi, S. Ikeda, H. Okamura, M. Ichinokawa, and S. Nakayama. "An asymmetric logistic regression model for ecological data". In: *Methods in Ecology and Evolution* 7.2 (Oct. 2015), pp. 249–260. DOI: 10.1111/2041-210X.12473.

[87] M. Uemura, K. S. Kawabata, S. Ikeda, and K. Maeda. "Variable Selection for Modeling the Absolute Magnitude at Maximum of Type Ia Supernovae". In: *Publications of the Astronomical Society of Japan* 67.3 (June 2015), 55(9pp). DOI: 10.1093/pasj/psv031. arXiv: 1504.01470 [astro-ph.SR].

[88] K. Watanabe and S. Ikeda. "Entropic Risk Minimization for Nonparametric Estimation of Mixing Distributions". In: *Machine Learning* 99.1 (Apr. 2015), pp. 119–136. DOI: 10.1007/s10994-014-5467-7.

[89] M. Honma, K. Akiyama, M. Uemura, and S. Ikeda. "Super-resolution imaging with radio interferometry using sparse modeling". In: *Publications of the Astronomical Society of Japan* 66.5 (Oct. 2014), 95(14pp). DOI: 10.1093/pasj/psu070. arXiv: 1407.2442 [astro-ph.IM].

[90] S. Ikeda, H. Odaka, M. Uemura, T. Takahashi, S. Watanabe, and S. Takeda. "Bin mode estimation methods for Compton camera imaging". In: *Nuclear Instruments and Methods in Physics Research A* 760 (Oct. 2014), pp. 46–56. DOI: 10.1016/j.nima.2014.05.081. arXiv: 1312.4291 [astro-ph.IM].

[91] S. Ikeda and H. Kono. "Phase retrieval from single biomolecule diffraction pattern". In: *Optics Express* 20.4 (Feb. 2012), pp. 3375–3387. DOI: 10.1364/OE.20.003375. arXiv: 1101.1442 [physics.data-an].

[92] M. D. McDonnell, S. Ikeda, and J. H. Manton. "An Introductory Review of Information Theory in the Context of Computational Neuroscience". In: *Biological Cybernetics* 105.1 (July 2011), pp. 55–70. DOI: 10.1007/s00422-011-0451-9. arXiv: 1107.2984 [cs.IT].

[93] S. Ikeda. "Combining binary machines for multi-class: Statistical model and parameter estimation". In: *Journal of Physics: Conference Series* 233 (Mar. 2010). International Workshop on Statistical-Mechanical Informatics 2010 (IW-SMI2010) 7-10 Mar. 2010, Kyoto, Japan, p. 012006. DOI: 10.1088/1742-6596/233/1/012006.

[94] S. Ikeda and J. H. Manton. "Capacity of a single spiking neuron". In: *Journal of Physics: Conference Series, International Workshop on Statistical-Mechanical Informatics 2009 (IW-SMI2009)* 197 (Sept. 2009), p. 012014. DOI: 10.1088/1742-6596/197/1/012014.

[95] S. Ikeda and J. H. Manton. "Capacity of a Single Spiking Neuron Channel". In: *Neural Computation* 21.6 (June 2009), pp. 1714–1748. DOI: 10.1162/neco.2009.05-08-792.

[96] M. Hamada and S. Ikeda. "Channel Estimation and Code Word Inference for Mobile Digital Satellite Broadcasting Reception". In: *IEICE trans. on Communications* E91-B.12 (Dec. 2008), pp. 3886–3898. DOI: 10.1093/ietcom/e91-b.12.3886.

[97] Y. Sakaguchi and S. Ikeda. "Motor Planning and Sparse Motor Command Representation". In: *Neurocomputing* 70.Issues 10-12 (June 2007), pp. 1748–1752. DOI: 10.1016/j.neucom.2006.10.120.

[98] S. Ikeda, T. Tanaka, and S. Amari. "Information Geometry for Turbo Decoding". In: *Systems and Computers in Japan* 36.1 (Jan. 2005), pp. 79–87. DOI: 10.1002/scj.10359.

[99] S. Ikeda, T. Tanaka, and S. Amari. "Stochastic Reasoning, Free Energy, and Information Geometry". In: *Neural Computation* 16.9 (Sept. 2004), pp. 1779–1810. DOI: 10.1162/0899766041336477.

[100] S. Ikeda, T. Tanaka, and S. Amari. "Information geometry of turbo and low-density parity-check codes". In: *IEEE Transactions on Information Theory* 50.6 (June 2004), pp. 1097–1114. DOI: 10.1109/TIT.2004.828072.

[101] F. Asano, S. Ikeda, M. Ogawa, H. Asoh, and N. Kitawaki. "Combined Approach of Array Processing and Independent Component Analysis for Blind Separation of Acoustic Signals". In: *IEEE transactions on Audio and Speech Processing* 11.3 (May 2003), pp. 204–215. DOI: 10.1109/TSA.2003.809191.

[102] N. Murata, S. Ikeda, and A. Ziehe. "An Approach to Blind Source Separation Based on Temporal Structure of Speech Signals". In: *Neurocomputing* 41.1-4 (Oct. 2001), pp. 1–24. DOI: 10.1016/S0925-2312(00)00345-3.

[103] S. Ikeda and K. Toyama. "Independent Component Analysis for Noisy Data –MEG data analysis". In: *Neural Networks* 13.10 (Dec. 2000), pp. 1063–1074. DOI: 10.1016/S0893-6080(00)00071-X.

[104] S. Ikeda. "Acceleration of the EM algorithm". In: *Systems and Computers in Japan* 31.2 (Feb. 2000), pp. 10–18. DOI: 10.1002/(SICI)1520-684X(200002)31:2<10::AID-SCJ2>3.0.CO;2-D.

Chapters in books

[1] N. Murata and S. Ikeda. "EM Algorithm in Neural Network Learning". In: *The EM Algorithm and Related Statistical Models*. Ed. by M. Watanabe and K. Yamaguchi. STATISTICS: A Dekker series of Textbooks and Monographs 170. New York, NY/Basel: Marcel Dekker, Inc., Oct. 2003. Chap. 8, pp. 95–126. ISBN: 0824747011.

[2] S. Amari, S. Ikeda, and H. Shimokawa. "Information Geometry and Mean Field Approximation: The α-projection Approach". In: *Advanced Mean Field Methods – Theory and Practice*. Ed. by M. Opper and D. Saad. Cambridge, MA: MIT Press, Feb. 2001. Chap. 16, pp. 241–257. ISBN: 0262150549.

[3] S. Ikeda. "ICA on Noisy Data: A Factor Analysis Approach". In: *Advances in Independent Component Analysis*. Ed. by M. Girolami. Springer-Verlag London Ltd., June 2000. Chap. 11, pp. 201–215. ISBN: 1852332638.

Articles in conference proceedings

[1] Y. Koga, M. Uemura, R. Sazaki, and S. Ikeda. "Classification of Galactic Transients with Missing Data". In: *Proc. ADASS (Astronomical Data Analysis Software and Systems) XXXII*. Ed. by S. Gaudet, D. Bohlender, S. Gwyn, A. Hincks, and P. Teuben. Vol. 538. APS conference series. Sept. 2025, 348(4pp). DOI: 10.26624/FHG01766.

[2] K. Kohno, A. Endo, Y. Tamura, A. Taniguchi, T. Takekoshi, S. Ikeda, N. Yoshida, et al. "Sub/millimeter-Wave Dual-Band Line Intensity Mapping Using the Terahertz Integral Field Units with Universal Nanotechnology (TIFUUN) for the Atacama Submillimeter Telescope Experiment (ASTE)". In: *Millimeter, Submillimeter, and Far-Infrared Detectors and Instrumentation for Astronomy XII*. Ed. by J. Zmuidzinas and J.-R. Gao. Vol. PC13102. (48 authors). International Society for Optics and Photonics. Proc. SPIE, Aug. 2024, PC1310209. DOI: 10.1117/12.3021109.

[3] A. Taniguchi, Y. Tamura, S. Ikeda, T. Takekoshi, R. Kawabe, K. Kohno, and T. Sakai. "Data science based efficient and automated spectroscopy for submillimeter single-dish telescopes". In: *2023 XXXVth General Assembly and Scientific Symposium of the International Union of Radio Science (URSI GASS)*. Aug. 2023, pp. 1–4. DOI: 10.23919/URSIGASS57860.2023.10265475.

[4] T. Nakazato, S. Ikeda, K. Akiyama, G. Kosugi, M. Yamaguchi, and M. Honma. "New synthesis imaging tool for ALMA based on the sparse modeling". In: *Proc. ADASS (Astronomical Data Analysis Software and Systems) XXVIII*. Maryland, U.S.A., Nov. 2018, O4.1.

[5] G. Kosugi, T. Nakazato, and S. Ikeda. "Qualification of Sparse Modeling Technique for radio interferometric imaging of ALMA". In: *Proc. ADASS (Astronomical Data Analysis Software and Systems) XXVIII*. Maryland, U.S.A., Nov. 2018, P12.9.

[6] M. Morii, S. Ikeda, and Y. Maeda. "Image reconstruction method for an X-ray telescope with an angular resolution booster". In: *Proc. ADASS (Astronomical Data Analysis Software and Systems) XXVIII*. Maryland, U.S.A., Nov. 2018, P12.11.

[7] M. Yamaguchi, K. Akiyama, A. Kataoka, M. Fukagawa, T. Tsukagoshi, T. Muto, S. Ikeda, et al. "Super-resolution imaging of the protoplanetary disk HD 142527 using sparse modeling". In: *Proc. ADASS (Astronomical Data Analysis Software and Systems) XXVIII*. (7th/10 authors). Maryland, U.S.A., Nov. 2018, P13.22.

[8] Y. Wakamatsu, K. Isogai, T. Morita, T. Kato, D. Nogami, M. Uemura, and S. Ikeda. "New method of eclipse mapping and an application to HT Cas in the 2017 superoutburst". In: *Proceedings of Science (The Golden Age of Cataclysmic Variables and Related Objects IV, 2017)*. Vol. 315. Proceedings of Science. Sept. 2018, 025(7pp). DOI: 10.22323/1.315.0025.

[9] Y. Kojima, S. Sako, R. Ohsawa, H. Takahashi, M. Doi, N. Kobayashi, T. Aoki, et al. "Evaluation of large pixel CMOS image sensors for the Tomo-e Gozen wide field camera". In: *Proc. SPIE*. Vol. 10709, High Energy, Optical, and Infrared Detectors for Astronomy VIII. (11th/39 authors). July 2018, 107091T. DOI: 10.1117/12.2311301.

[10] S. Sako, R. Ohsawa, H. Takahashi, Y. Kojima, M. Doi, N. Kobayashi, T. Aoki, et al. "The Tomo-e Gozen wide field CMOS camera for the Kiso Shmidt telescope". In: *Proc. SPIE*. Vol. 10702, Ground-based and Airborne Instrumentation for Astronomy VII. (11th/39 authors). July 2018, 107020J. DOI: 10.1117/12.2310049.

[11] R. Ohsawa, S. Sako, H. Takahashi, Y. Kikuchi, M. Doi, N. Kobayashi, T. Aoki, et al. "Development of a real-time data processing system for a prototype of the Tomo-e Gozen wide field CMOS camera". In: *Proc. SPIE*. Vol. 9913, Ground-based and Airborne Instrumentation for Astronomy III. (10th/43 authors). Aug. 2016, p. 991339. DOI: 10.1117/12.2231615.

[12] S. Sako, R. Osawa, H. Takahashi, Y. Kikuchi, M. Doi, N. Kobayashi, T. Aoki, et al. "Development of a prototype of the Tomo-e Gozen wide-field CMOS camera". In: *Proc. SPIE*. Vol. 9908, Ground-based and Airborne Instrumentation for Astronomy VI. (10th/43 authors). Aug. 2016, 99083P. DOI: 10.1117/12.2231259.

[13] Y. Ariki, T. Inamura, S. Ikeda, and J. Morimoto. "Sparsely Extracting Stored Movements to Construct Interfaces for Humanoid End-effector Control". In: *Proceedings of the 2015 IEEE Conference on Robotics and Biomimetics*. (Invited talk). Zhuhai, China, Dec. 2015, pp. 1816–1821. DOI: 10.1109/ROBIO.2015.7419036.

[14] S. Ikeda. "Sparsity and Information Processing". In: *Symposium MEIS2015: Mathematical Progress in Expressive Image Synthesis*. Ed. by H. Ochiai and Y. Dobashi. Vol. 64. MI Lecture Note. (Invited talk). Kyushu University. Sept. 2015.

[15] S. Ikeda, K. Hayashi, and T. Tanaka. "Channel capacity and achievable rates of peak power limited AWGNC, and their applications to adaptive modulation and coding". In: *Proceedings of 2014 International Symposium on Information Theory and its Applications (ISITA2014)*. Melbourne, Australia, Oct. 2014, pp. 590–594. arXiv: 1005.3889v2 [cs.IT].

[16] S. Ikeda, H. Odaka, M. Uemura, T. Takahashi, S. Watanabe, and S. Takeda. "Compton Camera Imaging". In: *Proceedings of 2013 Seventh International Conference on Sensing Technology (ICST2013)*. Wellington, New Zealand, Dec. 2013, pp. 674–677.

[17] S. Ikeda. "Optimization of probability measure and its applications in information theory". In: *Proceedings of The Sixth Workshop on Information Theoretic Methods in Science and Engineering*. (Invited talk). Tokyo, Japan, Aug. 2013, p. 49.

[18] K. Watanabe and S. Ikeda. "Rate-distortion function for gamma sources under absolute-log distortion measure". In: *Proceedings of 2013 IEEE International Symposium on Information Theory (ISIT2013)*. Istanbul, Turkey, July 2013, pp. 2557–2561. DOI: 10.1109/ISIT.2013.6620688.

[19] K. Watanabe and S. Ikeda. "Convex formulation for nonparametric estimation of mixing distribution". In: *Proceedings of The Fifth Workshop on Information Theoretic Methods in Science and Engineering*. Aug. 2012, pp. 36–39.

[20] S. Ikeda and H. Kono. "Sparse Phase Retrieval". In: *Proceedings of 4th Workshop on Signal Processing with Adaptive Sparse Structured Representations (SPARS2011)*. Edinburgh, U.K., June 2011, p. 106.

[21] S. Ikeda and Y. Sakaguchi. "Motor Planning as an Optimization of Command Representation". In: *Proceedings of 48th IEEE Conference on Decision and Control (CDC2009)*. Shanghai, China, Dec. 2009, pp. 4499–4504. DOI: 10.1109/CDC.2009.5399672.

[22] S. Ikeda and J. H. Manton. "Spiking Neuron Channel". In: *Proceedings of 2009 IEEE International Symposium on Information Theory (ISIT2009)*. Seoul, Korea, June 28–July 3, 2009, pp. 1589–1593. DOI: 10.1109/ISIT.2009.5205817.

[23] Y. Shiraishi, K. Fukumizu, and S. Ikeda. "A game theoretical analysis of combining classifiers for multi-class classification problems". In: *Proceedings of the international workshop on data-mining and statistical science (DMSS2007)*. Vol. A702. SIG-DMSM. Tokyo, Japan, Oct. 2007, pp. 149–160. DOI: 10.11517/jsaisigtwo.2007.DMSM-A702_12.

[24] M. Hamada and S. Ikeda. "Improving mobile reception of digital satellite broadcasting". In: *Proceeding of The 18th Annual IEEE International Symposium on Personal, Indoor and Mobile Radio Communications (PIMRC'07)*. IEEE. Athens, Greece, Sept. 2007, pp. 1–5. DOI: 10.1109/PIMRC.2007.4394266.

[25] Y. Sakaguchi and S. Ikeda. "Motor Planning and Sparse Motor Command Representation". In: *Processingds of Fifteenth Annual Computational Neuroscience Meeting (CNS∗2006)*. Edinburgh, U.K., July 2006, p. 89.

[26] S. Ikeda. "Sparse representation and piece-wise linear kernel". In: *Proceedings of Signal Processing with Adaptive Sparse Structured Representations (SPARS'05)*. Rennes, France, Nov. 2005, pp. 123–126.

[27] S. Ikeda. "Information geometry of turbo and LDPC codes". In: *Proceedings of 2005 IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP2005)*. Philadelphia, U.S.A., Mar. 2005. DOI: 10.1109/ICASSP.2005.1416482.

[28] S. Ikeda, T. Tanaka, and S. Amari. "Information Geometry of Loopy BP". In: *Supplementary Proceedings of ICANN/ICONIP2003*. Istanbul, Turkey, June 2003, pp. 54–57.

[29] S. Ikeda, T. Tanaka, and S. Amari. "Information Geometry of Turbo Codes". In: *Proceedings of 2002 IEEE International Symposium on Information Theory (ISIT2002)*. Lausanne, Switzerland, June 2002, pp. 114–119. ISBN: 0-7803-7501-7. DOI: 10.1109/ISIT.2002.1023386.

[30] S. Ikeda, T. Tanaka, and S. Amari. "Information Geometrical Framework for Analyzing Belief Propagation Decoder". In: *Advances in Neural Information Processing Systems*. Ed. by T. G. Dietterich, S. Becker, and Z. Ghahramani. Vol. 14. NIPS∗2001. Cambridge, MA. Vancouver, BC Canada: MIT Press, Apr. 2002, pp. 407–414. ISBN: 0262042088. URL: https://proceedings.neurips.cc/paper/2001/file/d7a84628c025d30f7b2c52c958767e76-Paper.pdf.

[31] T. Tanaka, S. Ikeda, and S. Amari. "Information-geometrical significance of sparsity in Gallager codes". In: *Advances in Neural Information Processing Systems*. Ed. by T. G. Dietterich, S. Becker, and Z. Ghahramani. Vol. 14. NIPS∗2001. Cambridge, MA. Vancouver, BC Canada: MIT Press, Apr. 2002, pp. 527–534. ISBN: 0262042088. URL: https://proceedings.neurips.cc/paper/2001/file/dc513ea4fbdaa7a14786ffdebc4ef64e-Paper.pdf.

[32] S. Ikeda, T. Tanaka, and S. Amari. "Belief propagation and turbo code: Information geometrical view". In: *Proceedings of 2001 International Conference on Neural Information Processing (ICONIP2001)*. Shanghai, China, Nov. 2001, pp. 41–46.

[33] F. Asano, S. Ikeda, M. Ogawa, H. Asoh, and N. Kitawaki. "Blind source separation in reflective sound fields". In: *Proceedings of 2001 IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP2001)*. Salt Lake City, U.S.A., May 2001. DOI: 10.1109/ICASSP.2001.940210.

[34] F. Asano, S. Ikeda, M. Ogawa, H. Asoh, and N. Kitawaki. "Blind source separation in reflective sound fields". In: *Proceedings of Workshop on Hands-Free Speech Communication 2001 (HSC2001)*. Kyoto, Japan, Apr. 2001, pp. 51–54.

[35] S. Kajihara, K. Toyama, S. Ikeda, N. Murata, T. Kobayashi, Y. Kida, and Y. Yoshida. "Separation of MEG Signal by Independent Component Analysis". In: *Proceedings of the 12th International Conference on Biomagnetism (Biomag2000)*. Helsinki, Finland, Aug. 2000.

[36] S. Ikeda and K. Toyama. "ICA for noisy neurobiological data". In: *Proceedings of International Joint Conference on Neural Networks 2000 (IJCNN2000)*. Como, Italy, July 2000. DOI: 10.1109/IJCNN.2000.860755.

[37] S. Ikeda. "Factor Analysis Preprocessing for ICA". In: *Proceedings of International Workshop on Independent Component Analysis and Blind Signal Separation (ICA2000)*. Helsinki, Finland, June 2000, pp. 327–332.

[38] F. Asano and S. Ikeda. "Evaluation and Real-time implementation of blind source separation system using time-delayed decorrelation". In: *Proceedings of International Workshop on Independent Component Analysis and Blind Signal Separation (ICA2000)*. Helsinki, Finland, June 2000, pp. 411–415.

[39] H. G. Okuno, S. Ikeda, and T. Nakatani. "Combining Independent Component Analysis and Sound Stream Segregation". In: *Proceedings of IJCAI-99 Workshop on Computational Auditory Scene Analysis*. Stockholm, Sweden, Aug. 1999, pp. 92–98.

[40] S. Ikeda, S. Amari, and H. Nakahara. "Convergence of the Wake-Sleep Algorithm". In: *Advances in Neural Information Processing Systems*. Ed. by M. S. Kearns, S. A. Solla, and D. A. Cohn. Vol. 11. NIPS∗98. Cambridge, MA. Denver, CO, USA: MIT Press, June 1999, pp. 239–245. ISBN: 0262112450. URL: https://proceedings.neurips.cc/paper/1998/file/0771fc6f0f4b1d7d1bb73bbbe14e0e31-Paper.pdf.

[41] S. Ikeda and N. Murata. "A method of ICA in time-frequency domain". In: *Proceedings of International Workshop on Independent Component Analysis and Blind Signal Separation (ICA'99)*. Aussios, France, Jan. 1999, pp. 365–371.

[42] S. Ikeda and N. Murata. "A method of blind separation based on temporal structure of signals". In: *Proceedings of 1998 International Conference on Neural Information Processing (ICONIP'98)*. Vol. 2. Kitakyushu, Japan, Oct. 1998, pp. 737–742.

[43] N. Murata and S. Ikeda. "A On-line Algorithm for Blind Source Separation on Speech Signals". In: *Proceedings of 1998 International Symposium on Nonlinear Theory and its Applications (NOLTA'98)*. Crans-Montana, Switzerland, Sept. 1998, pp. 923–926.

[44] S. Ikeda and N. Murata. "An Approach to Blind Source Separation of Speech Signals". In: *Proceedings of 1998 International Conference on Artificial Neural Networks (ICANN'98)*. Skövde, Sweden, Sept. 1998, pp. 761–766.

[45] S. Ikeda. "Acceleration of the EM algorithm". In: *Proceedings of 1997 International Symposium on Nonlinear Theory and its Applications (NOLTA'97)*. Hawaii, HI, USA., Dec. 1997, pp. 755–758.

[46] K. Nakano, K. Konishi, R. Ishiyama, and S. Ikeda. "A Self-Organizing System with Cell-Specialization". In: *Proceedings of 1997 IEEE International Conference on Evolutionary Computing (ICEC'97)*. Indianapolis, IN, USA., Apr. 1997, pp. 279–284. DOI: 10.1109/ICEC.1997.592315.

[47] S. Ikeda. "Construct The Structure of Stochastic Multilayer Perceptron Using The Model Search Method". In: *Proceedings of 1996 International Symposium on Nonlinear Theory and its Applications (NOLTA'96)*. Kochi, Japan, Oct. 1996, pp. 197–200.

[48] S. Ikeda and K. Nakano. "Estimate The Source Structure Through Communication". In: *Proceedings of 1995 IEEE International Conference on Neural Networks (ICNN'95)*. Perth, WA, Australia, Dec. 1995, pp. 2799–2802. DOI: 10.1109/ICNN.1995.488175.

[49] K. Nakano, H. Hiraki, and S. Ikeda. "A Learning Machine That Evolves". In: *Proceedings of 1995 IEEE International Conference on Evolutionary Computing (ICEC'95)*. Perth, WA, Australia, Dec. 1995, pp. 808–813. DOI: 10.1109/ICEC.1995.487490.

[50] S. Ikeda. "Construction of Phoneme Models — Model Search of Hidden Markov Models —". In: *Proceedings of International Workshop on Intelligent Signal Processing and Communication Systems (ISPACS'93)*. Sendai, Japan, Oct. 1993, pp. 82–87.

[51] S. Ikeda, K. Nakano, and Y. Sakaguchi. "A Robot Organizing Purposive Behavior by Itself". In: *Proceedings of International Joint Conference on Neural Networks (IJCNN'92)*. Vol. 1. Baltimore, MD, USA, June 1992, pp. 570–575. DOI: 10.1109/IJCNN.1992.287151.

Technical reports

[1] S. Pike, K. Ebisawa, S. Ikeda, M. Morii, M. Mizumoto, and E. Kusunoki. *Application of data science techniques to disentangle X-ray spectral variation of super-massive black holes*. Tech. rep. JAXA-RR-16-007. JAXA/ISAS, Mar. 2017. DOI: 10.20637/JAXA-RR-16-007/0007. arXiv: 1701.05386 [astro-ph.HE].

[2] N. Tominaga, T. Morokuma, M. Tanaka, N. Yasuda, H. Furusawa, M. Morii, S. Ikeda, et al. *Supernova candidates discovered with Subaru/Hyper Suprime-Cam*. Tech. rep. ATel#7927. (23 authors). The Astronomer's Telegram, Dec. 2015. URL: www.astronomerstelegram.org/?read=7927.

[3] M. Kawakita, S. Ikeda, and S. Eguchi. *A bridge between boosting and a kernel machine*. Tech. rep. 1006. The Institute of Statistical Mathematics, Research Memorandum, Sept. 2006. URL: www.ism.ac.jp/editsec/resmemo/resmemo-file/resm1006.htm.

[4] S. Ikeda. *Sparse representation and piece-wise linear kernel*. Tech. rep. 962. The Institute of Statistical Mathematics, Research Memorandum, Oct. 2005. URL: www.ism.ac.jp/editsec/resmemo/resmemo-file/resm962.htm.

Abstracts in conference proceedings

[1] S. Ikeda and J. H. Manton. "Capacity of a single neuron channel". In: *3rd Australian Workshop on Mathematical and Computational Neuroscience, NeuroEng 2008*. Melbourne, Australia, Nov. 2008, p. 13.

[2] Y. Sakaguchi and S. Ikeda. "Motor Planning Based on Sparse Command Representation". In: *Brain-inspired Information Technology (BrainIT2006)*. Kitakyushu, Japan, Sept. 2006, P1–8.

[3] M. Tanigawa, K. Yoshikawa, K. Toyama, Y. Ohtani, Y. Ejima, S. Ikeda, and S. Kajihara. "Motion Adaptation Depresses Perception of Contour from Motion and Magnetoencephalography (MEG) Responses in V2/3 and V5". In: *30th Annual meeting. Society for Nueroscience Abstracts (SfN2000)*. Vol. 26, Part 1. New Orleans, U.S.A., Nov. 2000, p. 670.

[4] S. Kajihara, K. Yoshikawa, N. Murata, S. Ikeda, M. Tanigawa, Y. Ohtani, Y. Ejima, and K. Toyama. "Application of independent component analysis (ICA) to magnetoencephalography". In: *29th Annual meeting. Society for Nueroscience Abstracts (SfN1999)*. Vol. 25, Part 2. Miami, FL, USA., Oct. 1999, p. 1930.

Software

[1] T. Nakazato and S. Ikeda. "PRIISM: Python module for Radio Interferometry Imaging with Sparse Modeling". In: *Astrophysics Source Code Library* (June 2020). Bibcode: 2020ascl.soft06002N.

[2] K. Akiyama, F. Tazaki, K. Moriyama, I. Cho, S. Ikeda, M. Sasada, H. Okino, and M. Honma. "SMILI: Sparse Modeling Imaging Library for Interferometry". In: *Astrophysics Source Code Library* (Apr. 2019). Bibcode: 2019ascl.soft04005A.

Patent

[1] S. Kajihara, S. Amari, K. Toyama, and S. Ikeda. "Biosignal measuring method and apparatus". Patent No. US 6544170 B1. Apr. 2003.